Uncertainty principles from majorization and entropy in the binary case

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Abstract

Making one quantum measurement more predictable can force another to become less so. Majorization captures this trade-off with a single list of probabilities that holds for every state and bounds many measures of uncertainty at once, among them the Shannon entropy and families of its generalizations. Here we examine what this universality gains and what it costs in the simplest setting: two measurements with two outcomes each. The optimal majorization limits can be read off from the ellipse of expectation values that quantum states can reach. Yet no state attains them unless the two measurements are compatible; for incompatible measurements, the entropy limits they imply are therefore never exact, except in two extreme cases. Even so, these limits outperform the standard Maassen–Uffink relation unless the measurements are nearly mutually unbiased, and they become almost exact as the measurements approach compatibility. We also find the optimal limit for noisy and biased detectors, show that uncertainty persists even when both measurements can be performed jointly, extend the results to systems of any dimension, and relate them to squeezed and minimum-uncertainty states of light.

Introduction

The uncertainty principle is one of the defining features of quantum mechanics [1–3]: there are pairs of measurements whose outcomes cannot both be predicted with certainty, however carefully the system is prepared. This preparation uncertainty is distinct from measurement uncertainty, which limits how accurately incompatible observables can be measured jointly [4]; we are concerned with the former. First formulated for the position and momentum of a particle, it has since become a quantitative tool of quantum information science. It underlies the security of quantum key distribution [5–7], provides experimentally accessible criteria for entanglement [6, 8, 9], and is closely related to, though distinct from, the incompatibility of quantum measurements [10, 11]. These applications call for formulations that apply to measurements with a finite number of outcomes and whose bounds do not depend on the state of the system.

Robertson's variance-based relation [2] does not meet these requirements: its lower bound depends on the state, it can vanish even for incompatible observables, and variances are not natural when the outcomes are mere labels [12]. Entropies of the outcome distributions provide a remedy [12–16]. The best known entropic uncertainty relation, due to Maassen and Uffink [17], bounds the sum of the Shannon entropies of the outcomes of two measurements by a quantity that depends only on the largest overlap between their eigenvectors. Entropic uncertainty relations have since grown into a broad field with applications throughout quantum information theory [18, 19]; they have been refined in various regimes [20, 21] and tested in experiments [22, 23].

Each such relation, however, constrains a single measure of uncertainty, while different tasks call for different measures: the Shannon entropy quantifies average information, the min-entropy governs guessing probabilities [24], the Rényi family interpolates between these and other extremes, and the Tsallis family generalizes the Shannon entropy in a different way. Majorization [25], the order that captures when one probability distribution is more disordered than another, makes it possible to treat all of them at once. It is a central tool of quantum information theory, where it governs, for instance, the transformations of entangled states [26] and, more generally, quantum resource theories [27, 28]. In the context of uncertainty, the idea is to find a single probability vector, independent of the state, that majorizes the outcome statistics of every state; every uncertainty measure that respects majorization then obeys a bound determined by this one vector. Depending on whether the two outcome distributions are combined by a direct sum or by a tensor product, such measures include the Shannon and Tsallis entropies or the Shannon and all Rényi entropies, among many others. This program was initiated by Partovi [29] and developed into universal uncertainty relations by Friedland, Gheorghiu and Gour [30] and by Puchała, Rudnicki and Życzkowski [31]; Rudnicki *et al.* subsequently introduced a direct-sum formulation that yields stronger entropic bounds [32]. Majorization uncertainty relations have since been extended to generalized measurements and quantum operations [33, 34], sharpened further [35–38], in particular by exploiting the lattice structure of majorization, and verified experimentally [38, 39].

The price of this universality is only partly understood. For a qubit measured in two bases—the setting of many experiments and, for mutually unbiased bases, of the BB84

protocol [5]—entropy-specific relations can be optimized: the optimal Shannon relation is known [40, 41], in closed form over part of the range and through a one-parameter minimization otherwise, and so are optimal relations for Rényi entropies [42, 43]. Earlier majorization bounds have been compared with the Maassen–Uffink relation and with the optimal Shannon bound [34, 44], and the optimal direct-sum bound for two qubit bases has been identified in the majorization lattice [36]. What is missing is a complete account of when such bounds are tight, why, and how this depends on the uncertainty measure. In the simplest setting, where both approaches can be carried out completely, what exactly does majorization gain, and what does it cost?

Here we answer this question for pairs of binary (two-outcome) measurements. Several ingredients are known. For two qubit bases the relevant majorization bounds are special cases of the general constructions of refs. [30–32]; the optimality of the direct-sum bound and the states that saturate its partial sums were found in ref. [36]; the expectation values that qubit states can reach are known to fill an ellipse [44–46]; and for unbiased unsharp qubit observables the direct-sum bound was evaluated in ref. [34]. We give a unified and self-contained treatment that derives both bounds, their optimality and the states that saturate each of their partial sums from this ellipse.

Our main finding is that, unless the measurements commute, no state attains the optimal majorization bound. Consequently, every bound it implies for a strictly Schur-concave measure of uncertainty, such as the Shannon entropy, is strict; only the Rényi entropies of order zero and infinity are bounded tightly. Even so, the direct-sum Shannon bound outperforms the Maassen–Uffink relation unless the measurement axes on the Bloch sphere are within about nine degrees of orthogonal, i.e., unless the measurements are nearly mutually unbiased [34], and its ratio to the exact minimum tends to one as the measurements approach compatibility, where the Maassen–Uffink relation is weakest. Comparing both types of majorization bound with the exact optimum across the whole Rényi family, we find that the direct-sum form fails at large orders. We also obtain the optimal direct-sum bound for arbitrary noisy and biased two-outcome qubit measurements, show that preparation uncertainty persists for jointly measurable pairs, extend the results for projective measurements to binary projective measurements in any finite dimension, and relate them to the minimum-uncertainty and squeezed states of quantum optics. All analytical results were also checked numerically.

Results

Majorization uncertainty relations

For $x \in \mathbb{R}^n$ let $x^{\downarrow}$ denote its entries in non-increasing order and $L_k(x) = \sum_{i=1}^{k} x_i^{\downarrow}$ its Lorenz partial sums, with $L_0(x) = 0$; the Lorenz curve of x is the piecewise-linear interpolation of the points $(k, L_k(x))$, $k = 0, \ldots, n$. For x, y with equal total, x is *majorized* by y, written $x \prec y$, if $L_k(x) \leq L_k(y)$ for all k [25]. A function Φ is Schur-concave if $x \prec y$ implies $\Phi(x) \geq \Phi(y)$, and *strictly* Schur-concave if, in addition, $\Phi(x) > \Phi(y)$ whenever $x \prec y$ and $x^{\downarrow} \neq y^{\downarrow}$. Every function $\Phi(x) = \sum_i \phi(x_i)$ with ϕ (strictly) concave is (strictly) Schur-concave [25].

The uncertainty measures used below are the Rényi [47] and Tsallis [48] entropies

$$H_\alpha(p) = \frac{\log_2 \sum_i p_i^\alpha}{1-\alpha}, \qquad T_\gamma(p) = \frac{1 - \sum_i p_i^\gamma}{\gamma - 1}, \tag{1}$$

with $\alpha \in [0, \infty]$ and $\gamma > 0$; the cases $\alpha \in \{0, 1, \infty\}$ and $\gamma = 1$ are defined by continuity, with the Shannon entropy $H = H_1$ as the common limit ($T_\gamma \to H \ln 2$ as $\gamma \to 1$). The orders $\alpha = \infty$ and $\alpha = \frac{1}{2}$ give the min-entropy $H_\infty(p) = -\log_2 \max_i p_i$ and the max-entropy $H_{1/2}$ [24], and H_0 is the logarithm of the number of non-zero entries. All of them are Schur-concave; H_α is strictly Schur-concave for $0 < \alpha < \infty$ and T_γ for all $\gamma > 0$, whereas H_0 and H_∞ are not.

Let $p(\rho)$ and $q(\rho)$ denote the outcome distributions of two measurements M and N on a state ρ. For measurements in orthonormal bases $\{|m_i\rangle\}$ and $\{|n_j\rangle\}$, the Maassen–Uffink (MU) relation [17] reads

$$H(p) + H(q) \;\geq\; -\log_2 c, \qquad c = \max_{i,j} |\langle m_i | n_j \rangle|^2. \tag{2}$$

It constrains one particular functional of the pair (p, q); a different uncertainty measure requires a different relation. A majorization uncertainty relation [29–32] instead consists of a single state-independent vector ω such that

$$p(\rho) \oplus q(\rho) \;\prec\; \omega \quad \text{or} \quad p(\rho) \otimes q(\rho) \;\prec\; \omega \qquad \forall \rho. \tag{3}$$

We call these the direct-sum and the tensor-product form, respectively. Every Schur-concave function Φ then inherits the bound $\Phi(p \oplus q) \geq \Phi(\omega)$ or $\Phi(p \otimes q) \geq \Phi(\omega)$. Sums of Shannon or Tsallis entropies are Schur-concave functions of $p \oplus q$, since, the totals of p and q being fixed, they are sums of a concave function over all entries; sums of Shannon or Rényi entropies are Schur-concave functions of $p \otimes q$, since $H_\alpha(p \otimes q) = H_\alpha(p) + H_\alpha(q)$. The direct-sum form also yields bounds for Rényi entropies of order $\alpha \leq 1$, by a separate argument (Corollary 1), but not for all orders (Proposition 2).

Restricted to non-negative, non-increasing vectors of fixed length and fixed total, $\prec$ is a partial order, and in fact a complete lattice [49, 50].

Definition 1 Let $\mathcal{A}$ be a set of non-negative vectors of common length and total. A vector ω is an *optimal majorization bound* for $\mathcal{A}$ if $a \prec \omega$ for all $a \in \mathcal{A}$, and $\omega \prec \omega'$ for every ω' with this property; i.e., $\omega = \sup \mathcal{A}$ in the majorization lattice.

Lemma 1 ([49, 50]) *Let $S_k = \sup_{a \in \mathcal{A}} L_k(a)$. Then $\sup \mathcal{A}$ is the vector whose Lorenz curve is the least concave majorant of $k \mapsto S_k$ (with $S_0 = 0$). In particular, if $\omega_k = S_k - S_{k-1}$ is already non-increasing, then $\omega = \sup \mathcal{A}$.*

In all cases treated below the second, simpler statement applies. The quantities S_k have a transparent meaning: S_k is the largest value that any state can give to a sum of k entries, i.e., to the summed probabilities of k outcomes. Proofs of all results below are given in Methods.

Binary measurements and their achievable statistics

A qubit state is $\rho = \frac{1}{2}(\mathbb{1} + \vec{r}\cdot\vec{\sigma})$ with $|\vec{r}| \le 1$. A projective two-outcome measurement is $M = \{P_\pm\}$ with $P_\pm = \frac{1}{2}(\mathbb{1} \pm \hat{m}\cdot\vec{\sigma}) = |m_\pm\rangle\langle m_\pm|$, $|\hat{m}| = 1$, and similarly $N = \{Q_\pm\}$ with axis $\hat{n}$ and eigenvectors $|n_\pm\rangle$. The outcome distributions are

$$p = \left(\tfrac{1+x}{2}, \tfrac{1-x}{2}\right), \qquad q = \left(\tfrac{1+y}{2}, \tfrac{1-y}{2}\right), \tag{4}$$

where $x = \hat{m}\cdot\vec{r}$ and $y = \hat{n}\cdot\vec{r}$. The overlaps are $|\langle m_\pm|n_\pm\rangle|^2 = \frac{1}{2}(1 + \hat{m}\cdot\hat{n})$ and $|\langle m_\pm|n_\mp\rangle|^2 = \frac{1}{2}(1 - \hat{m}\cdot\hat{n})$, hence

$$c = \tfrac{1}{2}\big(1 + |\hat{m}\cdot\hat{n}|\big) = \cos^2\tfrac{\theta}{2}, \qquad \cos\theta = |\hat{m}\cdot\hat{n}|, \tag{5}$$

where $\theta \in [0, \frac{\pi}{2}]$ is the (acute) Bloch angle between the two measurement axes. Relabeling the outcomes of N maps $\hat{n} \mapsto -\hat{n}$ and does not affect any majorization statement, so we may and shall assume $\hat{m}\cdot\hat{n} = \cos\theta \ge 0$. Note that $\frac{1}{2} \le c \le 1$ and $\sqrt{c} \ge 1/\sqrt{2} > 1 - \sqrt{c}$.

A general two-outcome POVM $\{E_0, E_1 = \mathbb{1} - E_0\}$ can always be written as

$$E_0 = \tfrac{1}{2}\big[(1+\mu)\mathbb{1} + \eta\,\hat{a}\cdot\vec{\sigma}\big], \qquad \eta \ge 0,\ |\mu| + \eta \le 1, \tag{6}$$

with $|\hat{a}| = 1$, the constraint being equivalent to $0 \le E_0 \le \mathbb{1}$. Here η is the sharpness and μ the bias; rank-one projective measurements correspond to $\eta = 1$, $\mu = 0$.

The statistics (4) depend on $\vec{r}$ only through (x, y).

Lemma 2 ([44–46]) *For non-parallel axes, the set $\{(\hat{m}\cdot\vec{r}, \hat{n}\cdot\vec{r}) : |\vec{r}| \le 1\}$ is the filled ellipse*

$$\mathcal{E}_\theta = \big\{(x, y) : x^2 - 2xy\cos\theta + y^2 \le \sin^2\theta\big\}. \tag{7}$$

Its boundary consists exactly of the images of the pure states whose Bloch vectors lie in $\mathrm{span}\{\hat{m}, \hat{n}\}$. *Moreover, for all $\alpha, \beta \in \mathbb{R}$,*

$$\max_{|\vec{r}|\le 1} (\alpha x + \beta y) = |\alpha\hat{m} + \beta\hat{n}|. \tag{8}$$

Equivalently, $(x, y) \in \mathcal{E}_\theta$ if and only if the covariance matrix of the observables $A = \hat{m}\cdot\vec{\sigma}$ and $B = \hat{n}\cdot\vec{\sigma}$,

$$\Gamma = \begin{pmatrix} 1 - x^2 & \cos\theta - xy \\ \cos\theta - xy & 1 - y^2 \end{pmatrix}, \tag{9}$$

is positive semidefinite, because $\det\Gamma = \sin^2\theta - (x^2 - 2xy\cos\theta + y^2)$. Its off-diagonal entry contains the anticommutator $\frac{1}{2}\{A, B\} = \cos\theta\,\mathbb{1}$, which does not depend on the state. The ellipse is a disk for mutually unbiased measurements ($\theta = \pi/2$) and degenerates into the segment $x = y$ as $\theta \to 0$ [46].

Consequently the achievable set $\wp(M, N) = \{p(\rho) \oplus q(\rho)\} \subset \mathbb{R}^4$ is the image of $\mathcal{E}_\theta$ under the affine map $(x, y) \mapsto \frac{1}{2}(1 + x, 1 - x, 1 + y, 1 - y)$: it is convex and compact,

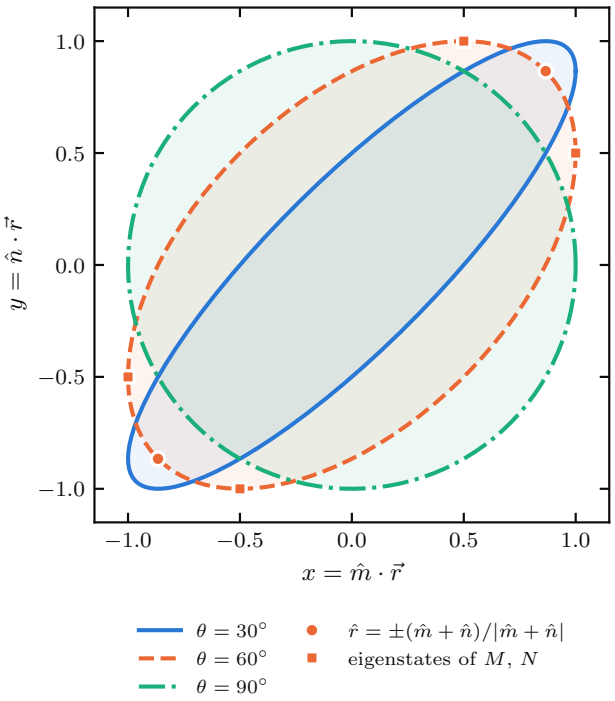


Fig. 1 Achievable expectation values for two qubit measurements. The ellipse $\mathcal{E}_\theta$ of achievable pairs $(x, y) = (\langle \hat{m} \cdot \vec{\sigma} \rangle, \langle \hat{n} \cdot \vec{\sigma} \rangle)$ for three Bloch angles. Markers (for $\theta = 60°$) show the two families of states that saturate the partial sums of $\omega_\oplus$: eigenstates of M or N (squares; $k = 1, 3$) and bisector states $\hat{r} \propto \pm(\hat{m} + \hat{n})$ (circles; $k = 2$). No single point saturates all partial sums (Proposition 1).

and its extreme points are the statistics of pure states in the plane of the two axes (Fig. 1). Any uncertainty measure that is concave in (x, y), such as $H(p) + H(q)$, is therefore minimized on $\partial\mathcal{E}_\theta$, and the optimization problems below are effectively one-dimensional.

Optimal bounds for two qubit bases

Theorem 1 *Let M, N be projective qubit measurements with maximal overlap c. For every state ρ,*

$$p(\rho) \oplus q(\rho) \;\prec\; \omega_\oplus := \big(1,\ \sqrt{c},\ 1 - \sqrt{c},\ 0\big), \tag{10}$$

and $\omega_\oplus$ is the optimal majorization bound for $\wp(M, N)$.

The majorization relation (10) is the qubit case of the direct-sum relation $p \oplus q \prec \{1\} \oplus W$ of Rudnicki *et al.* [32]: for $d = 2$ their vector $W = (s_1, s_2 - s_1)$ has $s_1 = \sqrt{c}$ and $s_2 = 1$. Its optimality, and the states that saturate its partial sums, were obtained by Li and Qiao in the majorization lattice [36]. In Methods we give a short proof based on Lemma 2.

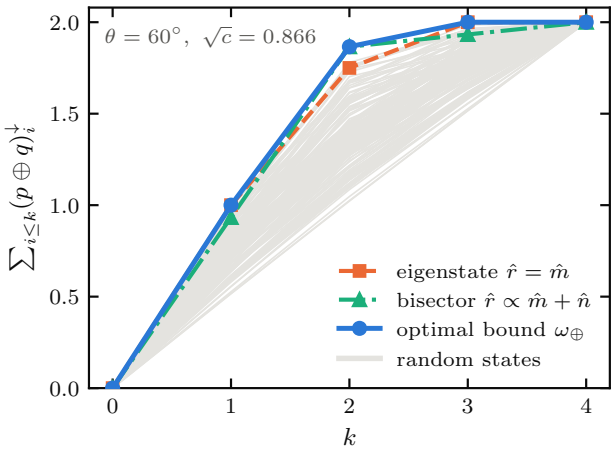


Fig. 2 The optimal bound as an upper envelope. Lorenz curves of $p \oplus q$ for $\theta = 60°$. Grey: 150 random states. The optimal bound $\omega_\oplus$ (blue) is the upper envelope; it touches the eigenstate curve at $k = 1, 3$ and the bisector-state curve at $k = 2$, but no single state reaches it. A similar plot for $\theta = 90°$ is Fig. 1(a) of ref. [36].

Proposition 1 *There is a state with $(p \oplus q)^{\downarrow} = \omega_\oplus$ if and only if $c = 1$, i.e., M and N commute. For $c < 1$, the maxima S_1, S_3 are attained only by the eigenstates of M and N, and S_2 only by the bisector states $\vec{r} = \pm(\hat{m}+\hat{n})/|\hat{m}+\hat{n}|$ (for $\theta = \pi/2$ also by $\vec{r} = \pm(\hat{m}-\hat{n})/|\hat{m}-\hat{n}|$).*

Proposition 1 makes precise the sense in which $\omega_\oplus$ is a *description* of the achievable set rather than the statistics of a particular state: it is the supremum of $\wp(M, N)$ in the majorization lattice, and its partial sums are saturated by two disjoint families of states (Fig. 2). The gap between $(1, c, 1-c, 0)$ and $\omega_\oplus$ quantifies the price one pays for demanding certainty about M.

Theorem 2 *For every state ρ,*

$$p(\rho) \otimes q(\rho) \ \prec\ \omega_\otimes := \big(w,\ 1-w,\ 0,\ 0\big), \tag{11}$$

where $w = \frac{1}{4}\big(1+\sqrt{c}\big)^2$, and $\omega_\otimes$ is the optimal majorization bound for $\{p(\rho) \otimes q(\rho)\}$.

The vector $\omega_\otimes$ coincides with the vector $Q^{(1)} = (C^2, 1-C^2, 0, 0)$, $C = (1+\sqrt{c})/2$, of Puchała *et al.* [31], which for $d = 2$ exhausts their hierarchy of bounds; see also ref. [30]. As for the direct sum, $\omega_\otimes$ is not attained by any state when $c < 1$ (Methods).

Entropic consequences and tightness

Let $h(t) = -t \log_2 t - (1-t)\log_2(1-t)$ be the binary entropy.

Corollary 1 *For every concave ϕ on $[0, 1]$, $\sum_i \phi(p_i) + \sum_j \phi(q_j) \ge \phi(1) + \phi(\sqrt{c}) + \phi(1 - \sqrt{c}) + \phi(0)$. In particular,*

$$H(p) + H(q) \ \ge\ h(\sqrt{c}), \tag{12}$$

$$T_\gamma(p) + T_\gamma(q) \;\geq\; \frac{1 - c^{\gamma/2} - (1-\sqrt{c})^\gamma}{\gamma - 1} \qquad (\gamma > 0), \tag{13}$$

and, for $0 < \alpha \leq 1$,

$$H_\alpha(p) + H_\alpha(q) \;\geq\; H_\alpha(\sqrt{c}, 1-\sqrt{c}), \tag{14}$$

where $H_\alpha(\sqrt{c}, 1-\sqrt{c}) = \log_2\big[c^{\alpha/2} + (1-\sqrt{c})^\alpha\big]/(1-\alpha)$.

Corollary 2 *For every* $\alpha \in [0, \infty]$,

$$H_\alpha(p) + H_\alpha(q) \;\geq\; H_\alpha(\omega_\otimes) = \frac{\log_2\big[w^\alpha + (1-w)^\alpha\big]}{1-\alpha}. \tag{15}$$

For $\alpha = 1$ *this reads* $H(p)+H(q) \geq h(w)$*; for* $\alpha = \infty$ *it gives* $H_\infty(p)+H_\infty(q) \geq -2\log_2 \frac{1+\sqrt{c}}{2}$*, which is tight (bisector state); since* $H \geq H_\infty$*, it contains Deutsch's Shannon bound [12], which has the same right-hand side.*

As noted above, the two forms are natural for different families: Tsallis entropies are additive over direct sums, as used in the proof of Corollary 1, whereas Rényi entropies are additive over tensor products, $H_\alpha(p \otimes q) = H_\alpha(p) + H_\alpha(q)$. Where both forms apply, the direct sum is stronger, but it cannot be extended to all orders:

Proposition 2 *(i) For* $0 < \alpha \leq 1$, $H_\alpha(\sqrt{c}, 1-\sqrt{c}) \geq H_\alpha(\omega_\otimes)$*, with equality iff* $c = 1$*; in particular* $h(\sqrt{c}) \geq h(w)$*. (ii) For* $c < 1$ *the direct-sum form* (14) *fails at* $\alpha = \infty$*:* $\min_\rho [H_\infty(p) + H_\infty(q)] = -2\log_2 \frac{1+\sqrt{c}}{2} < -\frac{1}{2}\log_2 c = H_\infty(\sqrt{c}, 1-\sqrt{c})$.

The decisive question is whether these bounds are tight.

Theorem 3 *Let* $c < 1$*. (i) For every continuous, strictly Schur-concave* Φ*,*

$$\min_\rho \Phi(p \oplus q) > \Phi(\omega_\oplus), \qquad \min_\rho \Phi(p \otimes q) > \Phi(\omega_\otimes). \tag{16}$$

Consequently the bounds (12)*,* (13) *and* (14)*, and the bounds* (15) *with* $0 < \alpha < \infty$*, are all strict. (ii) The bounds* (15) *of order* $\alpha = 0$ *and* $\alpha = \infty$ *are tight; they are attained by the eigenstates of* M *or* N *and by the bisector states, respectively.*

Theorem 3 makes precise in what sense the majorization bounds are optimal. As majorization bounds they cannot be improved (Theorems 1 and 2); yet, because the supremum is not the statistics of any state, the bound it induces is strict for every strictly Schur-concave measure. For such measures the optimum has to be found measure by measure, which is what entropy-specific relations do; only at the extreme orders $\alpha = 0$ and $\alpha = \infty$ does the majorization bound reach it.

Comparison with the Maassen–Uffink relation and with the exact optimum

Figure 3 compares the Shannon bounds with MU and with the exact minimum of $H(p)+H(q)$, which is known for qubits [40, 41]. By Lemma 2 the minimum lies on the

great circle through $\hat{m}$ and $\hat{n}$. For $\theta \le \bar{\theta}$ it is attained at the bisector states and equals $2h\big(\frac{1+\sqrt{c}}{2}\big)$ [41]. At $\bar{\theta}$ the bisector turns from a minimum into a saddle point along the circle; a second-derivative test shows that this happens when $\sqrt{\bar{c}}\,\operatorname{artanh}\sqrt{\bar{c}} = 1$, i.e., at $\bar{c} \approx 0.6948$ and $\bar{\theta} \approx 67.07°$. For $\bar{\theta} < \theta < \pi/2$ the minimizers bifurcate and the minimum has to be computed numerically [41]. As θ grows they move continuously towards the eigenstates of M and N, which they reach at $\theta = \pi/2$, where MU is tight.

The range in which the bisector is optimal depends on the uncertainty measure, as Zozor *et al.* showed for Rényi entropies [43]. The eigenstates are optimal for $\alpha \le \frac{1}{2}$, the bisector for $\alpha \ge \alpha^\star(c)$, and intermediate states in between, where $\alpha^\star$ decreases from $\alpha^\dagger \approx 1.43$ for mutually unbiased measurements to $\frac{1}{2}$ as $c \to 1$. In terms of the Bloch angle, the bisector is optimal up to 43° for $\alpha = 3/4$, up to $\bar{\theta} \approx 67°$ for the Shannon entropy and up to 80° for $\alpha = 1.2$, and at all angles for $\alpha \ge \alpha^\dagger$, in line with the tightness of the min-entropy bound. Here $\alpha^\dagger$ solves $2H_\alpha\big(\frac{2+\sqrt{2}}{4}, \frac{2-\sqrt{2}}{4}\big) = 1$, the condition for bisector and eigenstates to perform equally well when the measurements are mutually unbiased. For the max-entropy, for instance, the eigenstates give $\log_2(1+\sin\theta)$ and the bisector $2\log_2(1+\sin\frac{\theta}{2})$.

For mutually unbiased measurements ($\theta = 90°$, $c = \frac{1}{2}$) MU is tight (1 bit), while $h(1/\sqrt{2}) \approx 0.872$ and $h(w) \approx 0.844$. As the measurements become more compatible, however, the majorization bounds take over: $h(\sqrt{c}) > -\log_2 c$ for $\sqrt{c} > t^* \approx 0.7585$, i.e., for Bloch angles $\theta < \theta^* \approx 81.3°$ (the tensor bound overtakes MU at $\theta < 79.3°$), in agreement with ref. [34]; earlier majorization bounds [30, 31] were compared with MU and with the optimal qubit bound in ref. [44].

As $c \to 1$ the advantage over MU becomes unbounded, and the direct-sum bound becomes tight in ratio. With $\varepsilon = 1 - \sqrt{c} \approx \theta^2/8$ one has $-\log_2 c \approx 2\varepsilon/\ln 2$ and $h(\sqrt{c}) = h(\varepsilon) \approx \varepsilon\log_2(e/\varepsilon)$, whereas the exact minimum is $2h(\varepsilon/2) \approx \varepsilon\log_2(2e/\varepsilon)$. To leading order the ratio of the exact minimum to $h(\sqrt{c})$ is therefore $1 + 1/\log_2(e/\varepsilon)$: it tends to 1, but only logarithmically (1.09 at $\theta = 0.1$, 1.047 at $\theta = 0.003$), while the ratio to MU grows as $\frac{1}{2}\ln(16e/\theta^2)$ (4.2 and 7.7). The eigenstates of M, although they saturate S_1 and S_3, give $h(c)$, asymptotically twice the minimum.

Figure 4 extends the comparison to the whole Rényi family at $\theta = 60°$ and $\theta = 85°$. For $\alpha \le 1$ MU also applies, because H_α is non-increasing in α. At $\theta = 60°$ both majorization bounds beat MU for all $\alpha \le 1$; at $\theta = 85°$, closer to mutual unbiasedness, they do so only for $\alpha < 0.69$ (direct sum) and $\alpha < 0.58$ (tensor product). For the max-entropy, $\alpha = \frac{1}{2}$, the direct-sum bound reaches 83% and 91% of the exact minimum at the two angles. The expression $H_\alpha(\sqrt{c}, 1-\sqrt{c})$ remains a valid bound somewhat beyond $\alpha = 1$ in these examples, but it fails above $\alpha \approx 2.1$ ($\theta = 60°$) and $\alpha \approx 2.5$ ($\theta = 85°$), in line with Proposition 2(ii). For large α only the tensor-product bound remains, and it converges to the exact minimum as $\alpha \to \infty$. The majorization bounds thus become exact only at the two ends of the family, as Theorem 3 requires.

Finally, MU is one member of a family $H_\alpha(p) + H_\beta(q) \ge -\log_2 c$ with $1/\alpha + 1/\beta = 2$ [17], which includes the pair $H_\infty(p) + H_{1/2}(q) \ge -\log_2 c$ of min- and max-entropy. Majorization relations of the form (3) are symmetric under the exchange of p and q and therefore yield such mixed-order relations only indirectly, through the monotonicity of H_α in α.

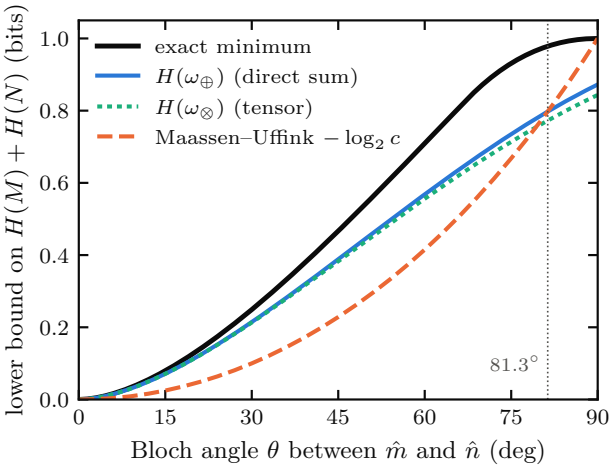


Fig. 3 Shannon-entropy bounds versus the Bloch angle. Lower bounds on $H(M) + H(N)$ as a function of the Bloch angle θ ($c = \cos^2\frac{\theta}{2}$). Black: exact minimum, equal to $2h\big(\frac{1+\sqrt{c}}{2}\big)$ for $\theta \le \bar{\theta} \approx 67.07°$. Blue: direct-sum bound $h(\sqrt{c})$. Green (dotted): tensor-product bound $h(w)$. Orange (dashed): Maassen–Uffink. The direct-sum bound is stronger than MU for $\theta < 81.3°$ (dotted vertical line).

Noisy and biased measurements

For POVMs, the partial sums of the direct-sum relation are largest eigenvalues of sums of effects, $S_k = \max_{|J|=k} \lambda_{\max}\big(\sum_{j\in J} F_j\big)$, where the F_j run over the effects of both measurements [34] (see also refs. [32, 33, 51]); Baek *et al.* evaluated them for unbiased unsharp qubit observables [34]. For two-outcome qubit POVMs the maximization can be carried out in closed form for arbitrary bias, and the resulting vector turns out to be ordered, hence optimal.

Let $M = \{E_0, E_1\}$ with parameters $(\mu, \eta, \hat{a})$ and $N = \{F_0, F_1\}$ with $(\nu, \zeta, \hat{b})$, as in equation (6). Then $p = \frac{1}{2}\big(1 + \mu + \eta\,\hat{a}\cdot\vec{r},\ 1 - \mu - \eta\,\hat{a}\cdot\vec{r}\big)$, and similarly for q.

Theorem 4 *Define*

$$\kappa = \max\{|\mu| + \eta,\ |\nu| + \zeta\}, \tag{17}$$

$$\Lambda = \max_{s,t=\pm1} \big[s\mu + t\nu + |s\eta\,\hat{a} + t\zeta\,\hat{b}|\big]. \tag{18}$$

Then $p(\rho) \oplus q(\rho) \prec \omega_\oplus$ *for all* ρ*, where*

$$\omega_\oplus = \tfrac{1}{2}\big(1 + \kappa,\ 1 + \Lambda - \kappa,\ 1 + \kappa - \Lambda,\ 1 - \kappa\big) \tag{19}$$

is the optimal majorization bound.

For projective measurements $\kappa = 1$, $\Lambda = 2\sqrt{c}$ and equation (19) reduces to equation (10). For unbiased measurements ($\mu = \nu = 0$), $\Lambda = (\eta^2 + \zeta^2 + 2\eta\zeta|\hat{a}\cdot\hat{b}|)^{1/2}$, in agreement with ref. [34]. If moreover $\eta = \zeta$, then $\omega_\oplus = \eta\,(1, \sqrt{c}, 1 - \sqrt{c}, 0) + \frac{1-\eta}{2}(1,1,1,1)$: such measurements are noisy versions of sharp ones [34, 52], and the achievable statistics, their supremum and the saturating states are those of the sharp

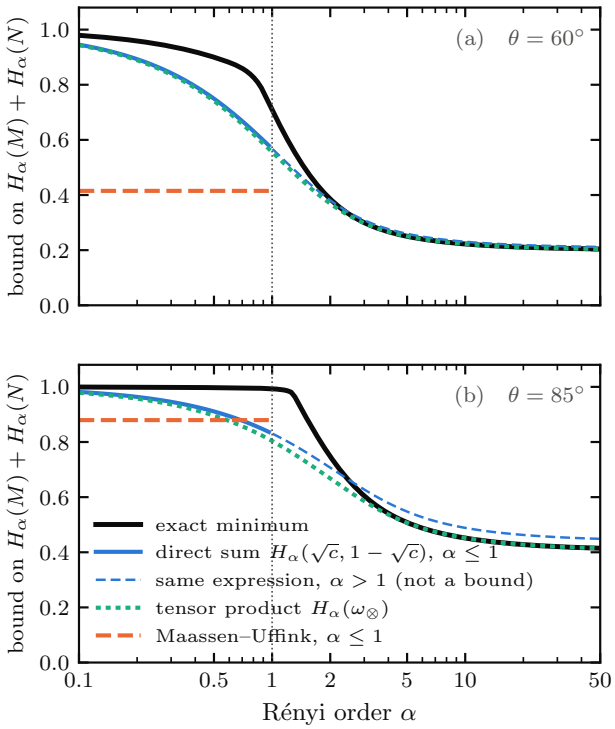


Fig. 4 Bounds across the Rényi family. Bounds on $H_\alpha(M) + H_\alpha(N)$ for (a) $\theta = 60°$ and (b) $\theta = 85°$. Black: exact minimum. Blue: direct-sum bound $H_\alpha(\sqrt{c}, 1-\sqrt{c})$ of Corollary 1, valid for $\alpha \le 1$; its thin dashed continuation for $\alpha > 1$ is not a bound and eventually exceeds the exact minimum. Green (dotted): tensor-product bound $H_\alpha(\omega_\otimes)$ of Corollary 2. Orange (dashed): MU, valid for $\alpha \le 1$. The dotted vertical line marks the Shannon case $\alpha = 1$.

case mixed with the uniform vector. For unequal sharpness or nonzero bias, $\omega_\oplus$ is not of this form, and the relevant geometric quantities are the lengths $|\eta\hat{a} \pm \zeta\hat{b}|$ rather than the overlap c.

Bias and unequal sharpness arise naturally in photon counting. If the two outputs of a polarizing beam splitter are monitored by detectors of efficiencies η_H and η_V, and no-click events are assigned to either outcome at random, the effect of outcome H has the form (6) with $\hat{a} = \hat{z}$, $\mu = (\eta_H - \eta_V)/2$ and $\eta = (\eta_H + \eta_V)/2$. A mismatch of detection efficiencies, a known concern in quantum key distribution [53], thus appears as a bias. A single click/no-click detector behind a polarizer, with no-click events counted as the second outcome, gives $\mu = \eta_d - 1$ and $\eta = \eta_d$, so that $|\mu| + \eta = 1$.

Uncertainty versus incompatibility.—For unbiased POVMs the vectors $\eta\hat{a} \pm \zeta\hat{b}$ that determine Λ also decide joint measurability: the pair is jointly measurable if and only if $|\eta\hat{a} + \zeta\hat{b}| + |\eta\hat{a} - \zeta\hat{b}| \le 2$ [10, 52]. The two notions are nevertheless distinct. For orthogonal axes and $\eta = \zeta \le 1/\sqrt{2}$ the measurements are jointly measurable, yet $S_2 = 1 + \eta/\sqrt{2}$ stays below the value $1 + \eta$ that would be reached if both outcome distributions could be sharpened at once, and $\omega_\oplus$ is not attained, being a noisy version of

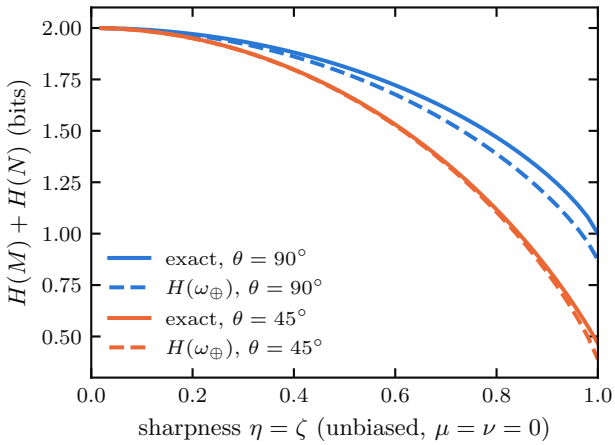


Fig. 5 Unsharp measurements. Exact minimum of $H(M) + H(N)$ (solid) and the bound from Theorem 4 (dashed) for unbiased measurements of equal sharpness ($\mu = \nu = 0$, $\eta = \zeta$) at Bloch angles 90° and 45°.

the sharp bound (Proposition 1). Majorization uncertainty relations therefore quantify preparation uncertainty, which persists in the absence of measurement incompatibility.

As a corollary, $H(p) + H(q) \geq -\sum_k \omega_k \log_2 \omega_k$ for all two-outcome qubit POVMs. Figure 5 shows that this bound tracks the exact minimum over the whole range of sharpness, becoming exact in the trivial limit $\eta \to 0$.

Binary projective measurements in any dimension

The qubit analysis already contains the general binary projective case.

Theorem 5 *Let $M = \{P, \mathbb{1} - P\}$ and $N = \{Q, \mathbb{1} - Q\}$ be projective measurements on $\mathbb{C}^d$ and $c = \max_{i,j} \|P_i Q_j\|_\infty^2$. Then Theorems 1 and 2 hold verbatim, and $\omega_\oplus$ is attained if and only if $c = 1$.*

In particular, all entropic bounds above hold for binary projective measurements in any dimension with $c = \max_{i,j} \|P_i Q_j\|^2$, and the min-entropy bound remains tight. Equivalently, $c = \frac{1}{2} + \frac{1}{4}\|\{A, B\}\|_\infty$ with $A = 2P - \mathbb{1}$ and $B = 2Q - \mathbb{1}$. Indeed, by Jordan's lemma (Methods), on a two-dimensional block, where P and Q act as qubit projectors with axes $\hat{m}_b$ and $\hat{n}_b$, $\{A, B\} = 2(\hat{m}_b \cdot \hat{n}_b)\mathbb{1}$ and $c_b = \frac{1}{2}(1 + |\hat{m}_b \cdot \hat{n}_b|)$, while $\{A, B\} = \pm 2$ on a one-dimensional block. For qubits this is the relation between the overlap and the effective anticommutator of ref. [44]. For arbitrary POVMs in any dimension, the eigenvalue formula for S_k quoted above, together with Lemma 1, gives an exact numerical procedure, which we use to verify Theorems 4 and 5 independently (Methods).

Discussion

The comparison with entropy-specific relations shows what the universality of majorization costs. Because the supremum is not the statistics of any state when the measurements do not commute, the induced bound is strict for every strictly Schur-concave measure, and it becomes exact only at the extreme orders $\alpha = 0$ and $\alpha = \infty$. Against a generic scalar relation such as MU, on the other hand, the majorization bounds are strong. For two projective qubit measurements, both Shannon bounds depend only on the Bloch angle θ between the two measurement axes: the direct-sum bound is $h(\sqrt{c})$ and MU gives $-\log_2 c$, with $c = \cos^2 \frac{\theta}{2}$. The two curves cross exactly once, at $\theta^* \approx 81.3°$ ($c \approx 0.575$), the root of $h(\cos \frac{\theta}{2}) = -2 \log_2 \cos \frac{\theta}{2}$ (Fig. 3). Nothing singles out this angle geometrically; it is simply where the two bounds exchange roles. For $\theta < \theta^*$, that is, unless the measurements are close to mutually unbiased, the direct-sum Shannon bound is the stronger of the two, and it becomes asymptotically tight in ratio as the measurements approach compatibility, precisely where MU is weakest. Only in the narrow window $\theta^* < \theta \leq 90°$ next to mutually unbiased measurements ($\theta = 90°$), for which MU is tight, does MU give the better bound; for the tensor-product bound the crossover lies at 79.3°. Among the two forms, the direct sum is preferable for $\alpha \leq 1$, while only the tensor product survives for large α.

These results contrast instructively with the quadratures of a light mode. There, coherent and squeezed states are minimum-uncertainty states: for their principal quadratures they saturate both the Heisenberg relation and the entropic relation of Białynicki-Birula and Mycielski [15]; moreover, all pure Gaussian states are equivalent under continuous majorization in phase space and are conjectured to majorize every Wigner-positive state [54]. For incompatible binary measurements no such state exists (Proposition 1), and the optimal state depends on the uncertainty measure. The ellipse $\mathcal{E}_\theta$ resembles the covariance ellipse of a squeezed vacuum: it is a disk for mutually unbiased measurements and flattens as they become compatible, although it describes the pair of measurements rather than a single state. Its boundary, traced by the pure states in the plane of the two axes, plays the role of the family of squeezed states, trading predictability of N for that of M. The balanced bisector state, the analogue of a coherent state, minimizes the Shannon sum up to $\bar{\theta} \approx 67°$; for larger angles the minimizer moves towards an eigenstate of either measurement, the analogue of an infinitely squeezed state, and the range in which the balanced state is optimal shrinks for smaller Rényi orders and grows for larger ones [43]. A single qubit cannot be squeezed in the usual sense [55], as all its pure states are spin coherent states; the essential difference is algebraic. For quadratures the commutator is a c-number, which makes the Heisenberg bound state independent. For binary observables it is the anticommutator that is, blockwise, a c-number [equation (9)], and it fixes $c = \frac{1}{2} + \frac{1}{4}\|\{A, B\}\|_\infty$ (see the remark after Theorem 5). The Robertson bound $\Delta A \, \Delta B \geq |\langle (\hat{m} \times \hat{n}) \cdot \vec{\sigma} \rangle|$, by contrast, vanishes for every state whose Bloch vector lies in the plane of the two axes, which contains all optimal states. Commutator-based relations are therefore uninformative here, and state-independent descriptions are needed: tight variance relations [46] or, for all Schur-concave measures at once, majorization.

The results apply directly to photonic qubits. Restricted to single photons, the Stokes operators of polarization are the Pauli matrices, and two linear polarization

analyzers at a relative angle ϑ correspond to $\theta = 2\vartheta$, so that $c = \cos^2\vartheta$ is Malus's law; the crossover with MU and the angle $\bar{\theta}$ correspond to $\vartheta \approx 40.7°$ and $\vartheta \approx 33.5°$, respectively. The optimal direct-sum bound has been tested in this setting with coherent light, the polarization state being reconstructed from measured Stokes parameters [39]. For many photons the Stokes operators form a collective spin that can be squeezed (polarization squeezing [56, 57]); the single-photon case is the limit in which only the redistribution of uncertainty along the boundary of $\mathcal{E}_\theta$ remains. Imperfect detectors lead to the unsharp and biased measurements of Theorem 4, with the bias set by the mismatch of detection efficiencies.

Several questions remain open. (i) *Tensor bounds for POVMs.* The arithmetic–geometric mean step in the proof of Theorem 2 is no longer tight for biased or unequally sharp measurements; the optimal $\omega_\otimes$ then requires maximizing products of affine functions over $\mathcal{E}_\theta$, and the least concave majorant in Lemma 1 may become non-trivial. (ii) *Binary POVMs in dimension d.* Jordan's lemma does not apply to pairs of effects directly. A Naimark dilation [58] turns them into projectors, but only product inputs are allowed, so Theorem 5 does not apply as it stands. (iii) *More measurements.* For three or more binary measurements the partial sums involve signed sums of several Bloch vectors, and the lattice structure of the bound becomes genuinely relevant. (iv) *Mixed orders.* It remains open whether majorization can be adapted to yield relations such as $H_\infty(p) + H_{1/2}(q) \geq -\log_2 c$. (v) *Binary measurements on light modes.* In infinite dimension $c = \sup_{i,j} \|P_i Q_j\|^2$ can equal 1 without a common eigenvector. For photon-number parity and displaced parity, whose expectation values sample the Wigner function [59], increasingly squeezed vacua, which contain only even photon numbers, approach simultaneous eigenstates of both, so no non-trivial state-independent relation exists. Tests for coherent states, $\{|\nu\rangle\langle\nu|, \mathbb{1} - |\nu\rangle\langle\nu|\}$, realized by a displacement followed by on/off detection with no-click probability $\pi Q(\nu) = \langle\nu|\rho|\nu\rangle$ (the Husimi function), also have $c = 1$, because all states orthogonal to both coherent states are common eigenvectors. Individual partial sums remain informative, however: $\pi[Q(\nu_1) + Q(\nu_2)] \leq 1 + e^{-|\nu_1-\nu_2|^2/2}$, with equality for an equal-weight superposition of $|\nu_1\rangle$ and $|\nu_2\rangle$, an optical cat state [60, 61], which thus plays the role of the bisector state. Non-trivial relations require $\|\{A, B\}\|_\infty < 2$, as for the pseudospin operators of ref. [62], of which the squeezed vacuum is an eigenstate, or for polarization measurements on single photons.

In summary, for binary measurements the majorization approach to uncertainty can be carried out completely. The achievable statistics form an affine image of an ellipse, and a single four-component vector, determined by the overlap c (or, for POVMs, by κ and Λ), is their least upper bound in the majorization lattice, for the direct sum as well as for the tensor product. This vector controls many uncertainty measures at once, but no state attains it when the measurements are incompatible, so the bounds it induces are never exact except at the extreme Rényi orders. The Shannon bound it implies is nevertheless stronger than the Maassen–Uffink relation for all but nearly mutually unbiased measurements, and it is asymptotically exact in ratio as the measurements approach compatibility.

Methods

Proofs

Lemma 1 is proven in refs. [49, 50]. The remaining results are proven here, in the order in which they are stated in Results.

Proof of Lemma 2. Write $\vec{r} = \vec{r}_\parallel + \vec{r}_\perp$ with $\vec{r}_\parallel \in \mathrm{span}\{\hat{m}, \hat{n}\}$. Only $\vec{r}_\parallel$ contributes to (x, y), and $(x, y) = G\,\xi$ where ξ are the coefficients of $\vec{r}_\parallel$ in the basis $\{\hat{m}, \hat{n}\}$ and $G = \left(\begin{smallmatrix} 1 & \cos\theta \\ \cos\theta & 1 \end{smallmatrix}\right)$ is the Gram matrix. Then $|\vec{r}_\parallel|^2 = \xi^T G \xi = (x, y) G^{-1} (x, y)^T \leq 1$, which is the stated inequality since $\det G = \sin^2\theta$. Equality holds iff $\vec{r}_\perp = 0$ and $|\vec{r}| = 1$. Equation (8) is Cauchy–Schwarz: $\alpha x + \beta y = \vec{r} \cdot (\alpha\hat{m} + \beta\hat{n})$. □

Proof of Theorem 1. We compute $S_k = \max_\rho L_k(p \oplus q)$ for $k = 1, \ldots, 4$. Clearly $S_1 \leq 1$ and $S_4 = 2$. Since all entries are non-negative, $S_3 = 2 - \min_\rho \min_i (p \oplus q)_i \leq 2$. Both bounds are attained by an eigenstate of M, for which $p = (1, 0)$. For $k = 2$, the two largest entries either belong to the same distribution, with sum 1, or are of the form $p_i + q_j$. By Lemma 2,

$$p_i + q_j = 1 + \tfrac{1}{2}(\pm x \pm y) \leq 1 + \tfrac{1}{2} \max\{|\hat{m} + \hat{n}|, |\hat{m} - \hat{n}|\}. \tag{20}$$

With $\hat{m} \cdot \hat{n} = \cos\theta \geq 0$ the maximum is $|\hat{m} + \hat{n}| = \sqrt{2(1 + \cos\theta)} = 2\cos\frac{\theta}{2} = 2\sqrt{c}$, so

$$S_2 = 1 + \sqrt{c}, \tag{21}$$

attained by the bisector state $\vec{r} = (\hat{m} + \hat{n})/|\hat{m} + \hat{n}|$. This is the qubit case of the Landau–Pollak-type inequality $p_i + q_j \leq 1 + \sqrt{c}$ underlying Deutsch's relation [12, 63]. The increments of $(S_1, S_2, S_3, S_4) = (1, 1 + \sqrt{c}, 2, 2)$ give $\omega_\oplus$, which is non-increasing because $1 \geq \sqrt{c} \geq 1 - \sqrt{c} \geq 0$. Lemma 1 completes the proof. □

Proof of Proposition 1. $(p \oplus q)^\downarrow_1 = 1$ forces ρ to be an eigenstate of M or N, say $\vec{r} = \pm\hat{m}$. Then $q = \left(\frac{1 \pm \cos\theta}{2}, \frac{1 \mp \cos\theta}{2}\right)$ and $(p \oplus q)^\downarrow = (1, c, 1 - c, 0)$, which equals $\omega_\oplus$ iff $c = \sqrt{c}$, i.e., $c = 1$. The statement about S_2 follows from the equality condition in Cauchy–Schwarz and from $|\hat{m} - \hat{n}| < |\hat{m} + \hat{n}|$ for $\theta < \pi/2$. □

Proof of Theorem 2. By the arithmetic–geometric mean inequality and equation (21), $p_i q_j \leq \left(\frac{p_i + q_j}{2}\right)^2 \leq \left(\frac{1 + \sqrt{c}}{2}\right)^2 = w$, with equality for the bisector state, for which $p_+ = q_+ = (1 + \sqrt{c})/2$. Hence $S_1 = w$. For an eigenstate of M, $p \otimes q = (q_+, q_-, 0, 0)$, so $S_2 = S_3 = S_4 = 1$. Since $\sqrt{c} \geq 1/\sqrt{2}$ we have $w \geq \frac{1}{2}$, so the increments $(w, 1 - w, 0, 0)$ are non-increasing, and Lemma 1 applies. Moreover, $\omega_\otimes$ is not attained for $c < 1$: equality in both steps of the argument forces a bisector state, at which $L_2(p \otimes q) = p_+ = (1 + \sqrt{c})/2 < 1$. □

Proof of Corollary 1. The first statement is Karamata's inequality applied to Theorem 1. Equations (12) and (13) follow with $\phi(t) = -t \log_2 t$ and $\phi_\gamma(t) = (t - t^\gamma)/(\gamma - 1)$, since $\sum_i p_i = \sum_j q_j = 1$ gives $T_\gamma(p) + T_\gamma(q) = \sum_k \phi_\gamma\big((p \oplus q)_k\big)$. Equation (14) is the qubit case of Theorem 2 of ref. [32]; we recall the short argument. For $\alpha < 1$ write $\sum_i p_i^\alpha = 1 + X$ and $\sum_j q_j^\alpha = 1 + Y$ with $X, Y \geq 0$. Then

$$H_\alpha(p) + H_\alpha(q) = \frac{\log_2[(1 + X)(1 + Y)]}{1 - \alpha}$$

$$\geq \frac{\log_2(1+X+Y)}{1-\alpha},$$

and $1+X+Y = \sum_k (p\oplus q)_k^\alpha - 1 \geq \sum_k (\omega_\oplus)_k^\alpha - 1 = c^{\alpha/2} + (1-\sqrt{c})^\alpha$ by Schur-concavity of $v \mapsto \sum_k v_k^\alpha$. □

Proof of Proposition 2. (i) Since $w - \sqrt{c} = \frac{1}{4}(1-\sqrt{c})^2 \geq 0$, we have $(\sqrt{c}, 1-\sqrt{c}) \prec (w, 1-w)$, with equality iff $c = 1$; apply strict Schur-concavity. The analogous statement in any dimension is proven in ref. [32]. (ii) With $t = \sqrt{c}$, $\left(\frac{1+t}{2}\right)^2 - t = \left(\frac{1-t}{2}\right)^2 > 0$ for $t < 1$; the minimum is the tight bound of Corollary 2. □

Proof of Theorem 3. (i) The achievable statistics form a compact set, so the minimum of Φ is attained at some state ρ^*. By Theorem 1 and Proposition 1, $p \oplus q(\rho^*) \prec \omega_\oplus$ with $(p\oplus q)^\downarrow \neq \omega_\oplus$, hence $\Phi(p\oplus q(\rho^*)) > \Phi(\omega_\oplus)$. The tensor product is analogous, using the non-attainment of $\omega_\otimes$ shown in the proof of Theorem 2. The Shannon and Tsallis sums are strictly Schur-concave functions of $p \oplus q$, and $H_\alpha(p) + H_\alpha(q) = H_\alpha(p\otimes q)$ with H_α strictly Schur-concave for $0 < \alpha < \infty$. For (14) with $\alpha < 1$, apply the argument to $v \mapsto \sum_k v_k^\alpha$ in the proof of Corollary 1. (ii) An eigenstate of M gives $H_0(p) + H_0(q) = 0 + 1 = H_0(\omega_\otimes)$, and a bisector state gives $\max_i p_i \max_j q_j = w$. □

Proof of Theorem 4. The largest single outcome probability is $\max_\rho \frac{1}{2}\big(1 + |\mu + \eta\, \hat{a}\cdot\vec{r}|\big) = \frac{1}{2}(1+|\mu|+\eta)$ for M, so $S_1 = \frac{1}{2}(1+\kappa)$; likewise the smallest is $\frac{1}{2}(1-\kappa)$, so $S_3 = 2 - \frac{1}{2}(1-\kappa)$. For S_2, a pair within one distribution sums to 1, while a cross pair gives, by equation (8),

$$\max_\rho \big[1 + \tfrac{1}{2}\big(s(\mu + \eta\,\hat{a}\cdot\vec{r}) + t(\nu + \zeta\,\hat{b}\cdot\vec{r})\big)\big]$$
$$= 1 + \tfrac{1}{2}\big[s\mu + t\nu + |s\eta\hat{a} + t\zeta\hat{b}|\big]. \quad (22)$$

Choosing $s = \operatorname{sgn}\mu$, $t = \operatorname{sgn}\nu$ shows $\Lambda \geq 0$, so $S_2 = 1 + \frac{1}{2}\Lambda$. The increments of (S_k) are the entries of equation (19). They are non-increasing iff $\kappa \leq \Lambda \leq 2\kappa$. The upper inequality follows from the triangle inequality, $\Lambda \leq |\mu| + |\nu| + \eta + \zeta \leq 2\kappa$. For the lower one, let ρ^* attain S_1 via an outcome of, say, M; then $S_2 \geq S_1 + \max_j q_j(\rho^*) \geq S_1 + \frac{1}{2}$, i.e., $\Lambda \geq \kappa$. Lemma 1 completes the proof. □

Proof of Theorem 5. By Jordan's lemma [64], which is also the standard tool for entropic uncertainty relations of binary measurements [44], $\mathbb{C}^d$ decomposes into mutually orthogonal subspaces $\mathcal{H}_b$ of dimension one or two, each invariant under P and Q. On a two-dimensional block, P and Q act as rank-one projectors, i.e., as a pair of qubit measurements with overlap c_b; on a one-dimensional block they have a common eigenvector, and we set $c_b = 1$. Since P and Q are block diagonal, $p(\rho)\oplus q(\rho) = \sum_b \lambda_b\, p_b \oplus q_b$ with $\lambda_b = \operatorname{tr}(\rho\Pi_b)$ and p_b, q_b the statistics of the normalized block state $\Pi_b\rho\Pi_b/\lambda_b$. Each $p_b\oplus q_b \prec \omega_\oplus(c_b) \prec \omega_\oplus(c)$, because $\omega_\oplus$ increases in the majorization order with c and $c = \max_b c_b$. Since $\{v : v \prec \omega\}$ is convex, $p\oplus q \prec \omega_\oplus(c)$; optimality follows by embedding the maximizing qubit states of a block with $c_b = c$. For the tensor product, $p_i q_j \leq (\frac{p_i+q_j}{2})^2$ with $p_i + q_j \leq \lambda_{\max}(P_i + Q_j) = 1 + \|P_i Q_j\| \leq 1 + \sqrt{c}$; $S_1 = w$ is attained in a block with $c_b = c$, and $S_2 = 1$ by any state in the range of P. Finally, if $c < 1$, a state with $L_1(p\oplus q) = 1$ lies in the range of some P_i (or Q_j); then $q_j = \operatorname{tr}(\rho P_i Q_j P_i) \leq \|P_i Q_j\|^2 \leq c$ for all j, so $L_2(p\oplus q) \leq 1 + c < 1 + \sqrt{c}$ and $\omega_\oplus$ is not attained. If $c = 1$, the ranges of some P_i and Q_j intersect and a common eigenvector attains $\omega_\oplus = (1,1,0,0)$. □

Numerical methods

All analytical results were checked numerically. We evaluated Lorenz curves, the lattice supremum via least concave majorants [49], the analytic bounds of Theorems 1, 2 and 4 and Corollaries 1–2, the eigenvalue characterization $S_k = \max_{|J|=k} \lambda_{\max}(\sum_{j\in J} F_j)$, and exact minima of Rényi and Tsallis entropy sums.

The closed forms were compared with (a) lattice suprema over 1.8×10^5 grid states (maximal deviation $< 10^{-4}$), and (b) the eigenvalue characterization for 200 random qubit POVM pairs and for 150 random pairs of projectors of rank $d/2$ in dimensions $d = 2, 4, 6$, all with $c < 1$ (deviation $< 10^{-14}$). Pairs with a common eigenvector, which always occur for odd d or unequal ranks, have $c = 1$ and were checked separately; the anticommutator form of c was checked for 300 random pairs of projectors of arbitrary rank in dimensions $d \le 8$. We also checked Propositions 1 and 2, Corollaries 1–2, Theorem 3, the critical angle $\bar{\theta}$, the motion of the Shannon minimizer beyond $\bar{\theta}$, the range of Rényi orders in which the bisector is optimal, the covariance form of Lemma 2, and the jointly measurable example and the detector model of the section on noisy and biased measurements. The statements about light modes in the Discussion were checked in a Fock space of dimension 600.

The exact entropy minima are obtained by a dense grid over the great circle in $\mathrm{span}\{\hat{m}, \hat{n}\}$ refined by Nelder–Mead. This reduction is justified by Lemma 2: the Shannon entropy is concave in (x, y) and is therefore minimized on the boundary of $\mathcal{E}_\theta$. For unbiased measurements, moving $\vec{r}$ outwards makes both p and q more ordered, so the same holds for every Rényi and Tsallis entropy.

Data availability. No experimental data were generated in this study. All numerical results and figures can be reproduced with the code described under Code availability.

Code availability. The code that verifies the analytical results and generates the figures is available at https://github.com/RuiZhu1/UncertaintyPrincipleofQubits under the MIT license.

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Acknowledgements. R. Z. and M. F. C. acknowledge support by the Quantum Science and Technology-National Science and Technology Major Project (Grant No. 2025ZD0301000), the National Key Research and Development Program of China (Grant No. 2023YFA1407100), the Guangdong Province Science, Technology Major Project (Future functional materials under extreme conditions – 2021B0301030005) and the National Natural Science Foundation of China (Grant No. 12574092). [TODO: add G. G.'s funding acknowledgement (ask Gilad)]

Author contributions. R. Z. carried out the analytical and numerical work and wrote the first draft of the manuscript. G. G. and M. F. C. supervised the project.

All authors discussed the results and revised the manuscript. [TODO: draft; confirm with all authors]

Competing interests. The authors declare no competing interests.